

A conditional real-axis model for the fast exponential Schrodinger example

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Status

This is a conditional reduction packet. It does not claim a full solution of the open question in Arnal–Siegl. The packet proves the limiting first-order model and records the exact localization dependency that would promote the result to a full real-axis asymptotic for the source Schrodinger operator.

Source and Open Question

The source is Antonio Arnal and Petr Siegl, “Resolvent estimates for one-dimensional Schrodinger operators with complex potentials”, arXiv:2203.15938, published in *Journal of Functional Analysis* 284 (2023), Article 109856 [1]. Section 7.3 considers

$$H = -\partial_x^2 + ie^{x^2}, \quad \text{Dom}(H) = W^{2,2}(\mathbb{R}) \cap \text{Dom}(e^{x^2}),$$

and gives the imaginary-axis asymptotic. It then states that the real-axis Theorem 4.2 is not applicable and leaves the critical region next to the real axis open.

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7.3. Fast growing potential. Let $H = -\partial_x^2 + ie^{x^2}$ with $\text{Dom}(H) = W^{2,2}(\mathbb{R}) \cap \text{Dom}(e^{x^2})$. Then

$$\Psi(ib) = 2^{-\frac{2}{3}} \|A_{1, \frac{\pi}{2}}^{-1}\| b^{-\frac{2}{3}} (\log b)^{-\frac{1}{3}} (1 + \mathcal{O}(b^{-\frac{1}{3}} (\log b)^{\frac{1}{3}})), \quad b \rightarrow +\infty, \quad (7.6)$$

which is as before illustrated in Fig. 7.1. Since the decay of $\Psi(\lambda)$ on the imaginary axis is faster than for any polynomial potential, the region for uniform boundedness of $\Psi(\lambda)$ adjacent to the imaginary axis is correspondingly wider. Note that Theorem 4.2 on the behaviour of $\Psi(\lambda)$ for $\lambda \in \mathbb{R}_+$ is not applicable in this case, see also Figure 7.1, and therefore the description of the critical region next to the real axis is currently an open question although [30, Eq. (5.5)] provides a clue as to what it may look like.

APPENDIX A GENERALISED AIRY OPERATOR

Idea of the Proof

Arnal–Siegl’s real-axis proof localizes the Fourier variable near the turning frequencies $\pm\sqrt{a}$, rescales by a parameter t_a satisfying $t_a V_2(t_a) = 2\sqrt{a}$, and obtains a first-order model $-\partial_x + V_2(t_a x)/V_2(t_a)$. For regularly varying V_2 , this model converges to $-\partial_x + |x|^\beta$.

For $V_2(x) = e^{x^2}$, the normalized model is instead

$$B_t = -\partial_x + e^{t^2(x^2-1)}.$$

The potential tends to zero on $(-1, 1)$ and to $+\infty$ outside that interval. Thus B_t^{-1} should converge to Volterra integration on $(-1, 1)$, with a one-sided boundary condition at 1. The Volterra norm on an interval of length 2 is $4/\pi$. This packet proves that model statement rigorously.

Conditional Dependencies

Conditional Dependency 1 (real-axis localization transfer). *Let $t_a > 0$ be determined by $t_a e^{t_a^2} = 2\sqrt{a}$, and let*

$$B_t = -\partial_x + e^{t^2(x^2-1)}$$

be realized in $L^2(\mathbb{R})$ with maximal domain

$$\text{Dom}(B_t) = \{u \in L^2(\mathbb{R}) : -u' + e^{t^2(x^2-1)}u \in L^2(\mathbb{R})\}.$$

The required transfer assertion is

$$\|(H - a)^{-1}\| = e^{-t_a^2} \|B_{t_a}^{-1}\| (1 + o(1)), \quad a \rightarrow +\infty.$$

Dependency 1 is exactly where the source proof does not apply: the regular-variation limit $V_2(tx)/V_2(t) \rightarrow |x|^\beta$ and the symbol estimate $|V_2^{(n)}(x)| \leq C_n(1 + V_2(x))\langle x \rangle^{-n}$ fail for $V_2(x) = e^{x^2}$. The rest of this packet is independent of that missing step.

The Model Theorem

Theorem 1 (fast exponential first-order model). *For $t > 0$, let*

$$B_t = -\partial_x + W_t(x), \quad W_t(x) = e^{t^2(x^2-1)}$$

on $L^2(\mathbb{R})$ with maximal domain. Then $0 \in \rho(B_t)$, and

$$\|B_t^{-1}\| \longrightarrow \frac{4}{\pi}, \quad t \rightarrow +\infty.$$

Proof. The source appendix proves the needed maximal-accretive framework for operators $-\partial_x + W$ with $\Re W \geq 0$ and confining real part. In the present real positive case $W_t(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, so B_t has compact resolvent and empty spectrum. Solving the first-order equation with the L^2 condition at $+\infty$ gives the inverse kernel

$$(B_t^{-1}f)(x) = \int_x^\infty K_t(x, y) f(y) dy, \quad K_t(x, y) = \mathbf{1}_{\{x < y\}} \exp\left(-\int_x^y W_t(s) ds\right).$$

Let

$$K_0(x, y) = \mathbf{1}_{\{-1 < x < y < 1\}},$$

and let T_0 be the corresponding integral operator. Pointwise, $K_t(x, y) \rightarrow K_0(x, y)$ away from the boundary points ± 1 . Indeed, on compact subintervals of $(-1, 1)$, $W_t \rightarrow 0$, while on any interval with positive distance from $(-1, 1)$, $W_t \rightarrow +\infty$ uniformly.

We now show convergence in Hilbert–Schmidt norm. Fix $R > 1$. On the finite square $[-R, R]^2$, dominated convergence gives

$$\iint_{[-R, R]^2} |K_t - K_0|^2 dx dy \rightarrow 0.$$

It remains to control the tails. Since $W_t(\pm R) = e^{t^2(R^2-1)} \rightarrow +\infty$, the regions $x, y > R$ and $x, y < -R$ satisfy

$$\iint_{\{R < x < y\}} K_t(x, y)^2 dx dy \leq \frac{1}{2} \int_R^\infty \frac{dx}{W_t(x)} \rightarrow 0,$$

and, by the same estimate after reflecting $x \mapsto -x$,

$$\iint_{\{x < y < -R\}} K_t(x, y)^2 dx dy \rightarrow 0.$$

For $x < -R < y < R$, the interval $[x, -R]$ lies where $W_t \geq W_t(R)$, hence

$$K_t(x, y)^2 \leq \exp\{-2(-R - x)W_t(R)\},$$

and integration over $x < -R$ and $-R < y < R$ gives a bound $R/W_t(R) \rightarrow 0$. The region $-R < x < R < y$ is identical, and the mixed region $x < -R < R < y$ is even smaller. Therefore $K_t \rightarrow K_0$ in $L^2(\mathbb{R}^2)$, and

$$\|B_t^{-1} - T_0\| \leq \|K_t - K_0\|_{L^2(\mathbb{R}^2)} \rightarrow 0.$$

It remains to compute $\|T_0\|$. Under the translation of $(-1, 1)$ to $(0, 2)$, T_0 becomes the Volterra operator $Vf(s) = \int_s^2 f(r) dr$. The non-zero singular values of the Volterra operator on an interval of length L are $2L/((2n+1)\pi)$, $n = 0, 1, \dots$, obtained by differentiating the eigenvalue equation for V^*V . With $L = 2$, the largest singular value is $4/\pi$. Hence $\|B_t^{-1}\| \rightarrow 4/\pi$. $\square$

Conditional Consequence for the Source Operator

Theorem 2 (conditional real-axis asymptotic). *Assume Dependency 1. Then, for*

$$H = -\partial_x^2 + ie^{x^2}, \quad t_a e^{t_a^2} = 2\sqrt{a},$$

one has

$$\|(H - a)^{-1}\| = \frac{4}{\pi} e^{-t_a^2} (1 + o(1)), \quad a \rightarrow +\infty.$$

Equivalently, since $t_a^2 = \frac{1}{2}W_0(8a)$,

$$\|(H - a)^{-1}\| = \frac{\sqrt{2}}{\pi} \sqrt{\frac{W_0(8a)}{a}} (1 + o(1)) = \frac{\sqrt{2}}{\pi} \frac{\sqrt{\log a}}{\sqrt{a}} (1 + o(1)).$$

Moreover, for every fixed $\eta > 0$, the resolvent is conditionally uniformly bounded in the region

$$0 \leq b < \left(\frac{\pi}{4} - \eta\right) e^{t_a^2}$$

next to the positive real axis.

Proof. The norm asymptotic follows immediately by combining Dependency 1 with Theorem 1. The Lambert W formula is obtained from

$$t_a e^{t_a^2} = 2\sqrt{a} \iff 2t_a^2 e^{2t_a^2} = 8a.$$

For the boundedness region, the triangle inequality gives

$$\|(H - a - ib)u\| \geq \|(H - a)u\| - b\|u\|.$$

The preceding asymptotic says that the lower bound for $H - a$ is

$$\|(H - a)u\| \geq \left(\frac{\pi}{4}e^{t_a^2} + o(e^{t_a^2})\right)\|u\|.$$

Thus any fixed margin below $(\pi/4)e^{t_a^2}$ gives a uniform resolvent bound. $\square$

Verification Notes

The script `code/model_kernel_check.py` discretizes the explicit kernel K_t and compares its largest singular value with $4/\pi$. A sample run in the sandbox environment produced ratios decreasing toward 1 for $t = 24, 36, 48, 64, 80$. This computation is not used in the proof and is not evidence for Dependency 1; it is only a sanity check for the model constant.

Novelty and Literature Check

The lightweight run indexes had no exact packet for arXiv:2203.15938. Web searches on 2026-07-18 for the arXiv id, the source title, the exact phrase “critical region next to the real axis”, and variants involving e^{x^2} , Arnal, Siegl, and Krejcirik found the published JFA metadata and Arnal’s 2023 thesis metadata, but no obvious later paper resolving this exact fast-exponential real-axis edge. Because the packet is conditional, this is recorded only as a bounded status check, not as a novelty claim for a full solution.

Limitations and Human Review Focus

The proved model theorem does not by itself answer the source question. The missing step is to prove Dependency 1. A reviewer should focus on whether the Fourier-side localization, partition of unity, and commutator estimates in the proof of Arnal–Siegl Theorem 4.2 can be replaced by estimates adapted to $V_2(x) = e^{x^2}$. The obstruction is concrete: the derivatives of e^{x^2} grow like polynomials times e^{x^2} , so the source’s symbol-class estimates are not available globally.

References

- [1] Antonio Arnal and Petr Siegl, *Resolvent estimates for one-dimensional Schrodinger operators with complex potentials*, Journal of Functional Analysis 284 (2023), no. 9, Article 109856; arXiv:2203.15938.